

SAFELAN: ADAPTIVE SECURE FILE SYSTEM FOR LAN ENVIRONMENT

Project ID: 25-26J-067

Project Proposal Report

Dias W P S - IT22892744

BSc. Special (Hons.) Degree in Information Technology

(Specialized in Cyber Security)

Department of Computer System Engineering

Sri Lanka Institute of Information Technology

August 2025

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Supervisor: Mr. Kanishka Yapa Abeywardena

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement of any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Student Name	Student ID	Signature
Dias W P S	IT22892744	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.



Signature of the supervisor

28.08.2025

Date



Signature of the co-supervisor

28.08.2025

Date

ABSTRACT

This proposal presents the design and development of a Blockchain-based logging system integrated into the SafeShare-LAN project to enhance the security and integrity of LAN-based file-sharing operations. [1] [2] Traditional log management systems often suffer from vulnerabilities, including tampering, unauthorized access, and a lack of transparency [3] [4]. To address these challenges, the proposed solution leverages blockchain's immutability, decentralization, and transparency, ensuring that both the sender and the receiver obtain verifiable, tamper-proof logs for every file-sharing transaction, thereby strengthening accountability and auditability [1] [4] [5].

The system will be implemented on a private blockchain network that records critical events, including file encryption, transfers, and access attempts, as immutable transactions [2] [3] . Each log entry will be cryptographically hashed and stored on the blockchain, preventing any unauthorized modifications [4] [5]. To enhance usability, the solution provides real-time log visibility through a secure web-based interface, enabling both the file owner and the recipient to view and verify logs immediately after an event occurs [3].

The novelty of this component lies in its dual-log distribution and independent cryptographic verification mechanism, which enables both the sender and receiver to validate transaction authenticity without relying on a centralized authority or third-party system [4] [5]. The expected outcomes include enhanced data integrity, non-repudiation, trustworthiness, and full traceability within secure file-sharing environments [2] [3].

Keywords: Blockchain, Secure Logging, SafeShare-LAN, File Sharing Security, Immutability, Accountability, tamper-proof, non-repudiation

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1 INTRODUCTION

1.1 Background & Literature Survey

LAN-based file-sharing systems remain a fundamental component of organizational IT infrastructures because of their high-speed performance, ease of deployment, and cost-effectiveness [6] [7]. These systems facilitate quick and direct data exchange between users within the same network, reducing dependency on internet-based solutions. Despite these advantages, traditional LAN environments rely heavily on centralized logging mechanisms, where all operational and security-related logs are stored in a single location or server [8]. While this simplifies management, it also introduces a critical single point of failure. Suppose this central server is compromised by an attacker or an insider with privileged access. In that case, the logs can be modified, deleted, or fabricated, severely undermining the trustworthiness of the entire system [9].

Such vulnerabilities present significant risks to accountability, compliance, and digital forensics. Logs serve as evidence in security investigations and regulatory audits, and their compromise can lead to legal non-compliance, financial losses, and reputational damage for organizations [10] [11]. Moreover, insider threat employees or administrators with malicious intent pose an even greater challenge, as they often possess the necessary access privileges to manipulate logs without detection [12] [9].

To address these security gaps, blockchain technology emerges as a transformative solution due to its foundational characteristics:

- **Immutability:** Once a log entry is recorded on the blockchain, it cannot be modified or deleted, ensuring permanent and tamper-proof storage [1] [4].
- **Decentralization:** By distributing logs across multiple nodes, blockchain eliminates dependency on a single centralized server, reducing the risk of data manipulation by a single compromised point [13] [14].

- **Transparency and Verifiability:** All authorized stakeholders can independently verify the authenticity of logs without relying on a central authority, ensuring trust and accountability in the system [15] [16].

Recent studies and practical implementations demonstrate the effectiveness of blockchain for audit trails, compliance reporting, and integrity assurance in distributed and cloud environments [17] [8]. For example, platforms like Hyperledger Fabric and Ethereum private networks have successfully provided secure logging and data provenance for enterprise-level applications [18] [9]. However, these existing frameworks often cater to large-scale, internet-based distributed systems rather than LAN-based environments, which require low-latency, lightweight, and optimized solutions to avoid impacting performance [19].

Additionally, most current implementations focus on immutable storage but do not provide mechanisms for real-time dual-party verification, a critical feature in secure file-sharing environments where both the sender and receiver must confirm transaction authenticity immediately [11] [15]. Without this feature, disputes and repudiation issues may still arise.

The proposed solution in this research bridges these gaps by introducing a blockchain-integrated log management module within Safe-LAN, specifically tailored for LAN environments. This system ensures immutable, decentralized, and verifiable log storage, coupled with smart contract-based real-time log distribution to both parties involved in the communication. This design ensures non-repudiation, traceability, and insider threat resilience, setting a new standard for secure file sharing within organizational LAN infrastructures.

1.2 Research Gap

Table 1: Research Gap

Application Reference	Immutable Logs	Real-Time Log Visibility	Dual-Party Access Verification	LAN optimization
Research [3]	✓	✗	✗	✗
Research [7]	✓	✗	✗	✗
Research [12]	✓	✓	✗	✗
Research [19]	✗	✓	✗	✗
Proposed System	✓	✓	✓	✓

1.3 Research Problem

Despite studies on secure logging mechanisms for distributed systems, there is a considerable gap in research on tamper-proof and verifiable logging solutions optimized for LAN-based file-sharing environments. Most existing secure logging approaches rely on centralized storage or internet-based blockchain frameworks, making them unsuitable for organizations that operate in isolated LAN environments without internet connectivity [12] [10] [8]. Centralized systems remain vulnerable to insider threats and log tampering, compromising accountability and forensic investigations [9].

Although blockchain technology has shown promise for audit trails and compliance in enterprise systems [17] [3] [5], existing implementations are often resource-heavy, internet-dependent, and lack real-time dual-party log visibility [19] [15] [11]. Furthermore, current solutions rarely provide lightweight, LAN-optimized designs that ensure performance efficiency without compromising security [13] [20].

This project aims to fill this gap by developing a permissioned blockchain-based logging system that enables real-time, tamper-proof log storage, cryptographic linking, and shared visibility for both the file owner and the recipient in a LAN environment. The solution will ensure non-repudiation, traceability, and transparency while maintaining low latency and cost-effectiveness, making it suitable for small to medium-sized enterprises

2 OBJECTIVES

2.1 Main Objectives

The main objective of this study is to design and implement a permissioned blockchain-based logging system that provides secure, immutable, and auditable logs for the SafeShare-LAN application. The system will ensure tamper-proof log storage, real-time dual-party log visibility, and cryptographic integrity to enhance accountability and forensic reliability in LAN-based file-sharing environments.

2.2 Specific Objectives

The proposed system aims to configure a private blockchain network within the LAN to store all file access and modification logs in a distributed and immutable manner. This ensures that no single point of failure exists and that logs cannot be altered or deleted without detection.

Another key objective is to implement cryptographic hashing and smart contracts for secure and automated log management. Each log entry will be hashed and cryptographically linked to the previous entry, and smart contracts will handle the creation and validation of logs automatically.

The system will also provide real-time dual-party visibility by introducing a secure, browser-based interface that allows both the sender and receiver to access and verify logs independently. This enhances transparency and non-repudiation, eliminating the reliance on a central authority for log verification.

Finally, the solution will be rigorously tested to ensure performance efficiency, scalability, and security in LAN conditions. This includes evaluating latency, resource utilization, and the robustness of cryptographic mechanisms to maintain integrity without compromising speed or usability.

3 BLOCKCHAIN CONCEPTS AND THEIR RELEVANCE

3.1 Overview of Blockchain Technology

Blockchain is a decentralized, distributed ledger system designed to store data in a secure, transparent, and immutable manner [1], [4], [5]. Unlike traditional centralized logging systems, which rely on a single database or server, blockchain distributes copies of the ledger across multiple nodes. Each record (transaction) added to the blockchain is verified by network participants through a consensus mechanism, ensuring its integrity and authenticity before inclusion in the ledger.

In the context of LAN-based file-sharing systems, blockchain acts as an auditable trust layer for critical operations like file access and modifications, addressing issues of tampering, deletion, and insider threats that exist in centralized logging systems [3], [13], [11].

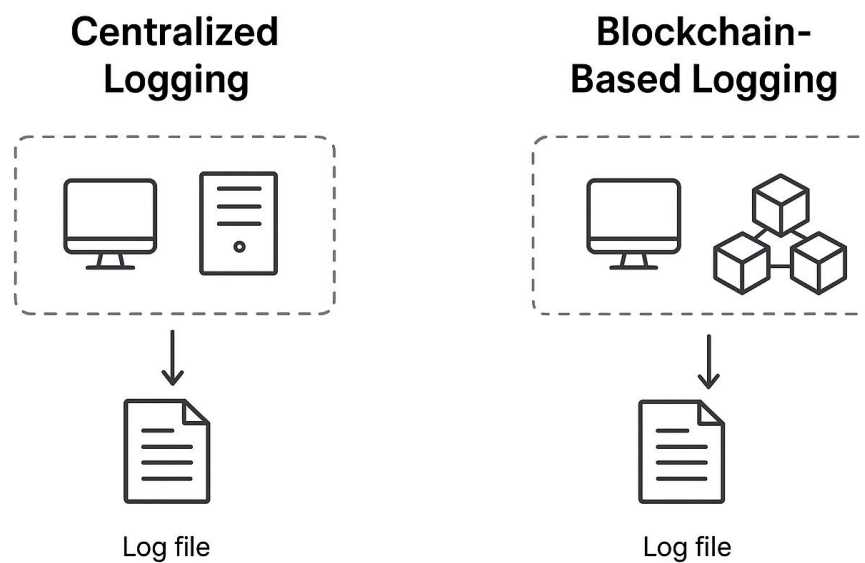


Figure 3-0-I : Comparison of centralized and blockchain-based logging for LAN file-sharing systems.

3.2 Core Components of Blockchain Technology

3.2.1 Blocks and Transactions

A blockchain consists of a chain of blocks, each containing:

- **Block Header:** Includes the previous block hash, timestamp, and Merkle root.
- **Block Body:** Contains transactions; in this case, file access logs such as File ID, User ID, timestamp, and operation type.

Transactions represent discrete events (e.g., file encryption, file transfer, or modification), which are validated and stored permanently on the ledger [4], [6]

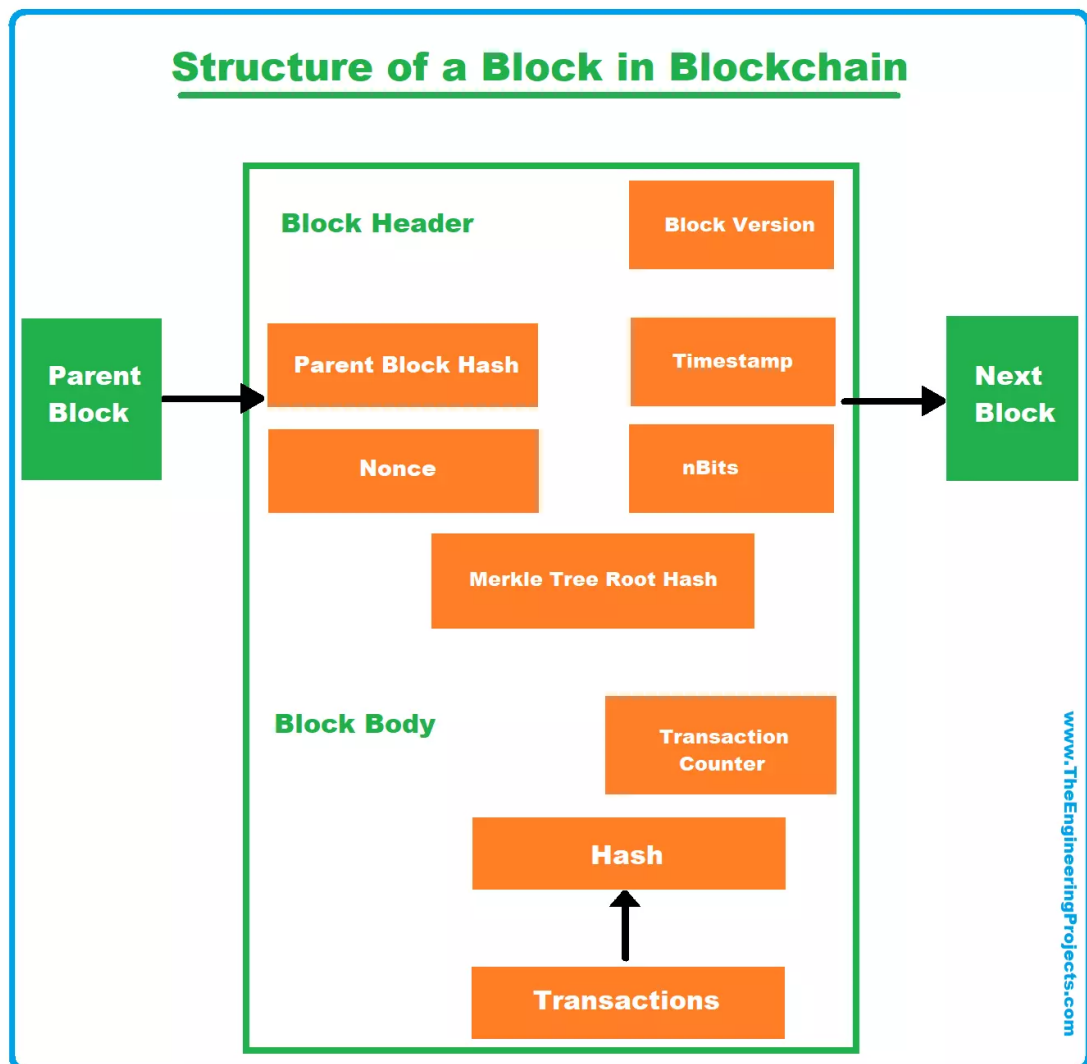


Figure 3-0-II: Block structure diagram

3.2.2 Cryptographic Hashing

Blockchain employs cryptographic hash functions like SHA-256 to generate a unique fixed-length hash for every block. Any modification to a transaction changes the hash value, invalidating subsequent blocks, thus ensuring immutability [5].

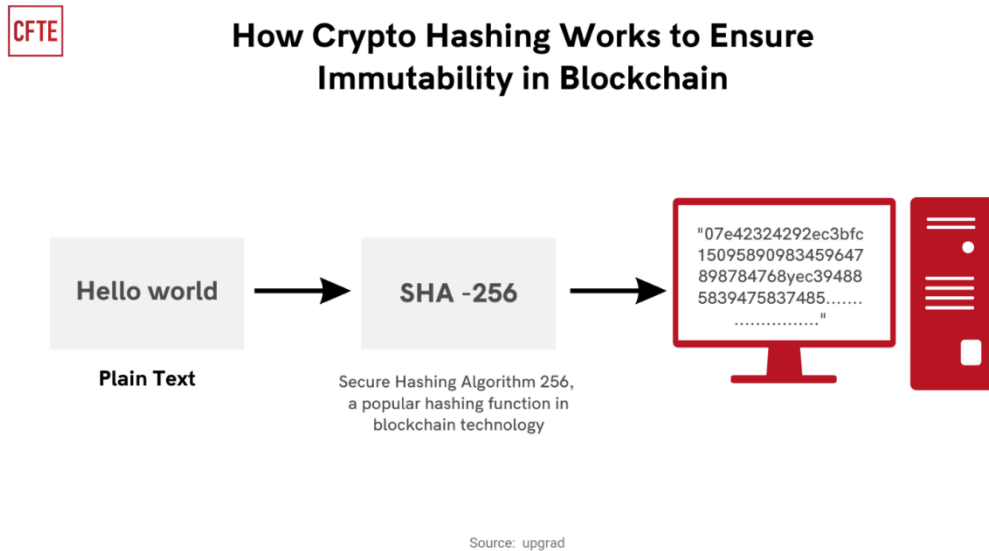


Figure 3-0-III: Cryptographic hashing ensures immutability of blockchain logs.

3.2.3 Consensus Mechanism

Consensus algorithms validate transactions and ensure all nodes agree on the state of the ledger. For LAN environments, Proof-of-Authority (PoA) or Raft consensus is preferred over energy-intensive Proof-of-Work (PoW) due to:

- Faster transaction validation
- Lower computational overhead
- Suitability for private, permissioned networks

PoA relies on pre-approved validator nodes, making it efficient for organizations where node participants are known and trusted [13], [14]

3.2.4 Smart Contracts

Smart contracts are self-executing programs deployed on the blockchain to automate operations. In the proposed system:

- Smart contracts will create and broadcast log entries when events occur (e.g., file access, download, or modification).
- They will distribute log hashes to both sender and receiver in real time for independent verification.
- This eliminates reliance on a third-party authority for log validation [3], [15].

3.3 Advantages of Blockchain for Secure Logging

Compared to traditional logging systems, blockchain offers the following advantages:

- **Immutability** – Once a log is written, it cannot be deleted or altered, ensuring tamper-proof evidence for audits [1] [4]
- **Transparency** – Authorized parties can view logs in real time without a central administrator.
- **Decentralization** – Removes the single point of failure associated with centralized servers.
- **Non-Repudiation** – Cryptographic signatures prevent users from denying actions logged on the blockchain [20] [11]

These benefits are critical for compliance with auditing standards and organizational security policies.

4 METHODOLOGY

The proposed component introduces a permissioned blockchain-based logging system within the Safe-LAN application to provide secure, immutable, and auditable logs for all file access and modification events in a LAN environment [1] [3] [12]. This methodology leverages the core properties of blockchain immutability, decentralization, and transparency to overcome the limitations of traditional centralized log management systems, which are vulnerable to tampering and insider threats [5] [8] [9].

The system will follow a structured approach, starting from the deployment of a permissioned blockchain network within the LAN, followed by the design of the log schema, implementation of smart contracts, and integration with the SafeShare-LAN application [13] [14] [20]. The methodology also incorporates performance testing to ensure that the blockchain integration does not introduce significant overhead while maintaining security and reliability [4] [19].

4.1 System Architecture

The architecture consists of the following layers:

- Application Layer (SafeShare-LAN) – Handles file sharing, encryption, and user authentication [6] [10].
- Blockchain Layer – A permissioned blockchain network (Hyperledger Fabric or Ethereum Private Network) that stores logs as tamper-proof transactions distributed across multiple nodes [18] [16].
- Smart Contract Layer – Implements automated rules for log creation, validation, and dual-party notification [3].
- User Interface Layer – A secure web dashboard providing real-time log visibility to both sender and receiver [17] [7].

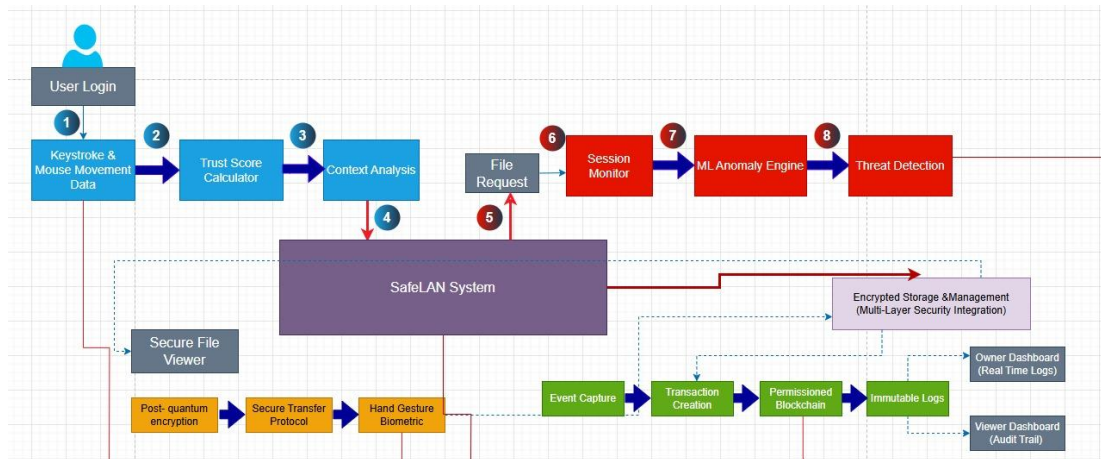


Figure 4-I: System Architecture

When a file-sharing event occurs (upload, download, modification), the SafeShare-LAN application sends log details such as File Hash, Sender ID, Receiver ID, Timestamp, and Event Type to the smart contract, which validates and records the transaction in the blockchain [1] [3]. Logs are cryptographically linked using SHA-256 hashing and stored in blocks, ensuring immutability [9] [11]. Both the owner and receiver are notified via the secure dashboard to verify the log entry in real time [2] [15]

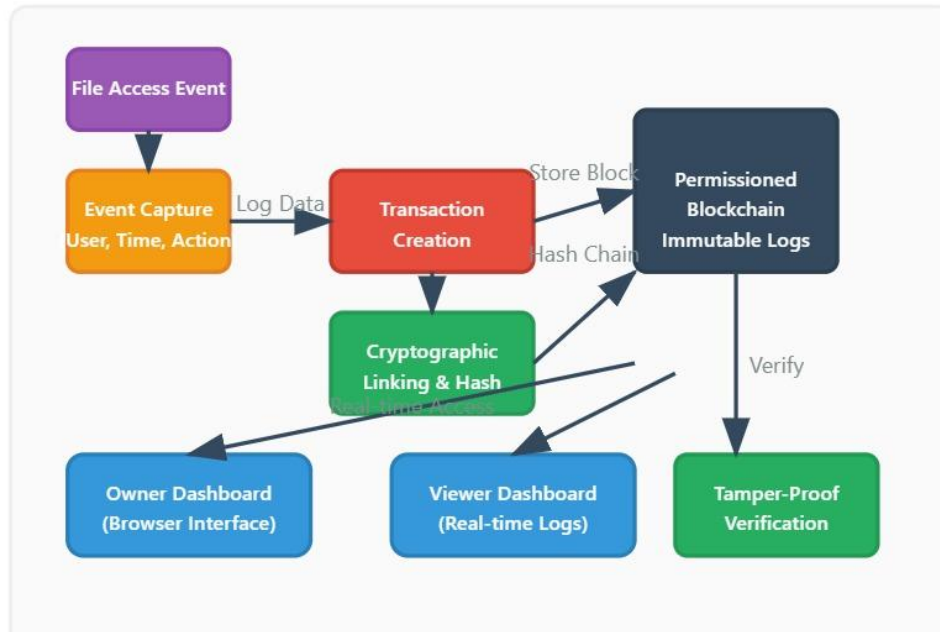


Figure 4-II:Blockchain process

4.2 Blockchain Integration and Consensus Mechanism

The blockchain used in this solution is a permissioned network, meaning only authorized nodes within the LAN can participate. This design ensures controlled access, privacy, and scalability compared to public blockchains [13] [14].

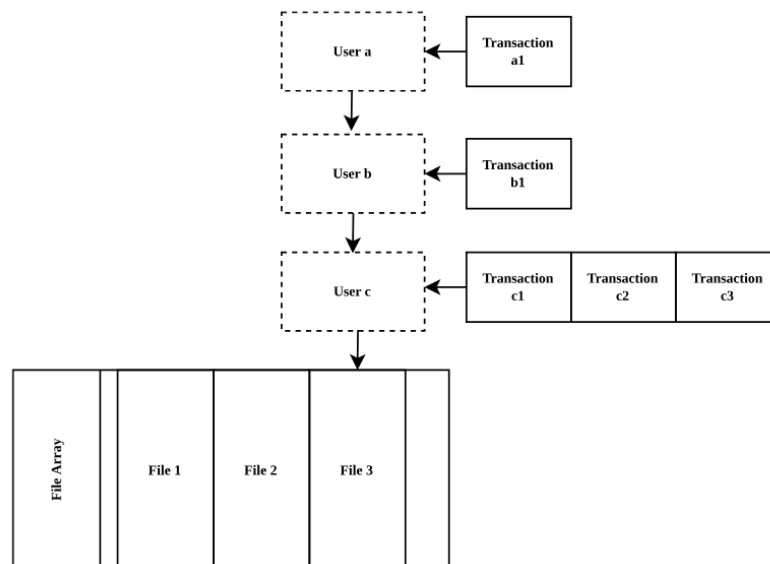


Figure 4-III: data structure of the File Transaction Chain

- Consensus Mechanism: RAFT or Proof of Authority (PoA) is used to achieve fast consensus with minimal computational overhead, making it ideal for LAN environments where nodes are trusted [4].
- Security Features:
 - SHA-256 Hashing ensures that log entries cannot be altered without invalidating subsequent blocks [9].
 - Digital Signatures (ECDSA) authenticate transactions, ensuring only authorized parties can write logs [15].
 - Merkle Trees provide efficient verification of log integrity [11].

4.3 Key Steps

Blockchain Setup

A permissioned blockchain will be deployed within the LAN to store all log data. Each node in the network will maintain a copy of the distributed ledger, eliminate single points of failure, and ensure redundancy [5].

Log Design and Schema

The log structure will include fields such as:

- Transaction ID
- File Hash (SHA-256)
- Timestamp
- Sender ID and Receiver ID
- Action Type (Upload/Modify/Delete)

These details will be stored as immutable blockchain transactions, cryptographically linked to the integrity [3] [13].

Smart Contract Development

Smart contracts will automate the process of log recording and real-time distribution. When a file action occurs, the contract will validate the transaction, store it in the ledger, and notify both parties via the web interface [11] [15].

Integration with SafeShare-LAN

The Safe-LAN application will interact with the blockchain through web3.py APIs. The API will allow the application to submit transactions, retrieve logs, and verify integrity without requiring manual intervention [20].

Real-Time Dual-Party Log Visibility

A secure web dashboard will allow both sender and receiver to view and verify logs immediately after each transaction, ensuring non-repudiation and shared accountability [11].

Testing and Evaluation

The system will undergo comprehensive testing for:

- Performance: Transaction latency, throughput, and scalability in LAN conditions [16].
- Security: Resistance to tampering, replay attacks, and insider threats [9].
- Novelty Verification: Comparing dual-party visibility and forensic reliability against traditional logging methods [20].

5 PROJECT REQUIREMENTS

5.1 Functional Requirements

- The system should maintain an immutable log of all file-sharing activities within the LAN environment.
- The system should store logs on a permissioned blockchain network to ensure tamper-proof integrity.
- The system should record details for each transaction, including timestamp, file hash, sender ID, receiver ID, and action type (upload, download, modify, delete).
- The system should enable real-time log visibility for both the file owner and the recipient through a secure web interface.
- The system should implement smart contracts to automate log creation, validation, and distribution.
- The system should allow both parties to verify the authenticity of logs independently without relying on a central authority.
- The system should integrate seamlessly with the SafeShare-LAN application for logging file-sharing activities.
- The system should notify both parties immediately after a log is created to ensure transparency and accountability.

5.2 Non-Functional Requirements

- **Security** – The system must use cryptographic hashing (SHA-256) and digital signatures to secure log entries. Access to the logs should require authentication and authorization to prevent unauthorized viewing.
- **User-Friendliness** – The web interface for log viewing should be intuitive, responsive, and easy to navigate for both sender and receiver. It should provide real-time updates and clear log details without complexity.
- **Reliability** – Logs must be permanently stored on the blockchain without any risk of tampering or deletion. The system should ensure high availability of the log data under normal LAN conditions.
- **Performance** – The system should maintain low latency for log recording and retrieval to avoid performance degradation during file-sharing operations. Blockchain transactions should be optimized for LAN speed.
- **Scalability** – The solution should support multiple users and nodes without significant performance loss. The blockchain should handle growing log data efficiently.
- **Transparency and Auditability** – Logs should be easily verifiable and auditable by authorized stakeholders for compliance and forensic investigations.

5.3 User Requirements

This system will cater to different user roles:

File Owners (Senders):

- Can view and verify blockchain-based logs of all files shared.
- Receive real-time notifications when a file access or modification occurs.

Recipients (File Viewers):

- Can independently verify log entries related to files they receive.
- Have transparent visibility of actions performed on shared files.

System Administrator:

- Manages blockchain nodes, user roles, and access control policies.
- Ensures the system operates efficiently and securely within the LAN environment.

5.4 Additional Requirements (Novelty Support)

- Dual-Party Log Visibility: Both the sender and recipient can view and validate logs without depending on a central authority.
- Tamper-Proof Audit Trails: All logs are immutable, ensuring forensic reliability for investigations.
- Offline LAN Operation: The system works entirely within the LAN without requiring internet connectivity.

6 GANTT CHART

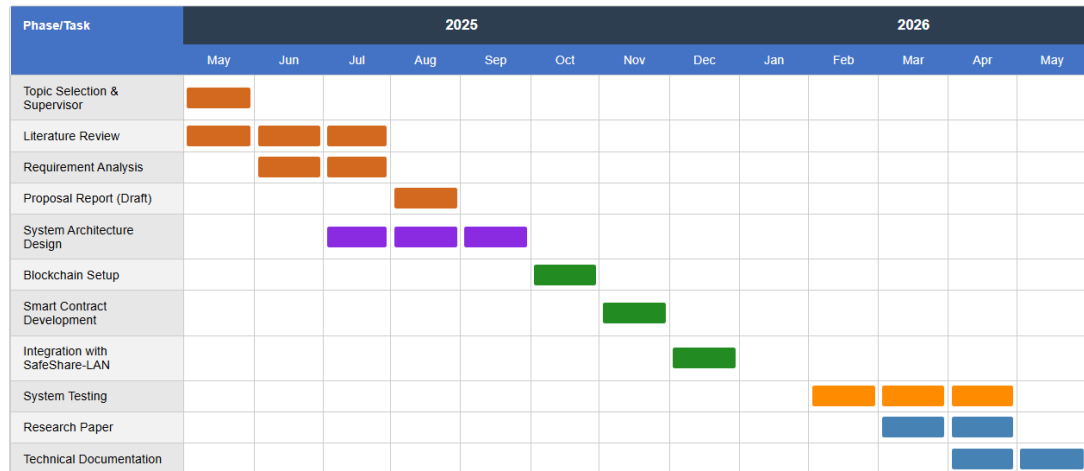


Figure 0-I: Gantt Chart

7 BUDGET AND BUDGET JUSTIFICATION

ITEM	DESCRIPTION	COST(LKR)
Blockchain tools	Open source	0
Development	-	0
Testing	LAN setup, multiple virtual nodes, and performance evaluation tools	10000
Miscellaneous Expenses	Printing, documentation, and power backup during testing	2000
TOTAL		12000/=

The proposed budget of LKR 12,000 is justified based on the nature of the project and its reliance on open-source technologies. Since blockchain tools such as Hyperledger Fabric and Ganache are open source, there is no cost for software licensing, and the development work will be completed in-house by the project team, resulting in zero development expenses. The major allocation of LKR 10,000 is for testing, which includes setting up a controlled LAN environment, deploying multiple virtual blockchain nodes, and utilizing performance evaluation tools to ensure the system meets scalability and security requirements. Additionally, LKR 2,000 has been allocated for miscellaneous expenses such as printing, documentation, and power backup during prolonged testing phases to prevent data loss or interruptions. This cost-effective budget ensures the successful implementation and evaluation of the blockchain-based logging system without unnecessary expenditure.

8 COMMERCIALIZATION ASPECTS

The proposed Blockchain-based Logging System for LAN-based file sharing can be commercialized as an enterprise security solution targeting organizations that require secure file sharing and immutable logging without internet dependency. Below are the key commercialization factors:

Market Opportunity

The solution targets small to medium enterprises (SMEs), educational institutions, financial organizations, and government offices that rely on LAN-based file sharing. With increasing compliance requirements and security threats, there is a strong demand for tamper-proof logging solutions in internal networks.

Business Model

- One-time licensing fee for the core system (SafeShare-LAN with Blockchain Module).
- Annual maintenance contracts (AMC) for updates and support.
- Enterprise customization services for integration with existing infrastructure.

Regulatory Compliance

- ISO 27001 for information security management.
- GDPR for data protection where applicable.
- Local government cybersecurity regulations for log retention and integrity.

Go-to-Market Plan

- Target Industries: SMEs, universities, hospitals, finance, and defense sectors.
- Promotion Strategies: Cybersecurity conferences, webinars, LinkedIn campaigns, partnerships with IT solution providers.
- Pilot Programs: Offer trial deployments in academic institutions and small organizations to demonstrate value.

Customer Support

Provide 24/7 technical support, training sessions for IT staff, and comprehensive user documentation. Offer remote troubleshooting and software updates as part of the AMC.

Continuous Improvement

Collect user feedback, integrate advanced analytics dashboards, and explore AI-powered anomaly detection in blockchain logs for predictive security insights. Regular updates will ensure compatibility with evolving LAN security standards

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10 TURNITIN

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